SPARQL 1.2 Overview

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Abstract

This document is an overview of SPARQL 1.1. It provides an introduction to a set of W3C specifications that facilitate querying and manipulating RDF graph content on the Web or in an

RDF store.

Status of This Document

This section describes the status of this document at the time of its publication. A list of current W3C publications and the latest revision of this technical report can be found in the W3C standards and drafts index.

This specification is published by the RDF Star Working Group as part of the update of specifications for format and errata.

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This document is governed by the 18 August 2025 W3C Process Document.

§ Set of Documents

This document is one of twelve SPARQL 1.2 documents produced by the RDF & SPARQL Working Group.

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§ 1. Introduction

SPARQL 1.1 is a set of specifications that provide languages and protocols to query and manipulate RDF graph content on the Web or in an RDF store. The standard comprises the following specifications:

- SPARQL 1.1 Query Language - A query language for RDF.
- SPARQL 1.1 Query Results JSON Format and SPARQL 1.1 Query Results CSV and TSV Formats - Apart from the standard SPARQL Query Results XML Format [SPARQL-XML-Result], SPARQL 1.1 now allows three alternative popular formats to exchange answers to SPARQL queries, namely JSON, CSV (comma separated values) and TSV (tab separated values) which are described in these two documents.
- SPARQL 1.1 Federated Query - A specification defining an extension of the SPARQL 1.1 Query Language for executing queries distributed over different SPARQL endpoints.
- SPARQL 1.1 Entailment Regimes - A specification defining the semantics of SPARQL queries under entailment regimes such as RDF Schema, OWL, or RIF.
- SPARQL 1.1 Update Language - An update language for RDF graphs.
- SPARQL 1.1 Protocol for RDF - A protocol defining means for conveying arbitrary SPARQL queries and update requests to a SPARQL service.
- SPARQL 1.1 Service Description - A specification defining a method for discovering and a vocabulary for describing SPARQL services.
- SPARQL 1.1 Graph Store HTTP Protocol - As opposed to the full SPARQL protocol, this specification defines minimal means for managing RDF graph content directly via common HTTP operations.
- SPARQL 1.1 Test Cases - A suite of tests, helpful for understanding corner cases in the specification and assessing whether a system is SPARQL 1.1 conformant

§ 1.1 Example

In the following, we will illustrate the use of SPARQL's languages, protocols, and related specifications with a small example.

Some RDF graph published on the Web at the URL 'http://example.org/alice' contains personal information about Alice and her social contacts. We use Turtle [Turtle] syntax here for illustration.

Graph: /http://example.org/alice

```
@prefix foaf: <http://xmlns.com/foaf/0.1/> .
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .

<http://example.org/alice#me> a foaf:Person .
<http://example.org/alice#me> foaf:name "Alice" .
<http://example.org/alice#me> foaf:mbox <mailto:alice@example.org> .
<http://example.org/alice#me> foaf:knows <http://example.org/bob#me> .
<http://example.org/bob#me> foaf:knows <http://example.org/alice#me> .
<http://example.org/bob#me> foaf:name "Bob" .
<http://example.org/alice#me> foaf:knows <http://example.org/charlie#m
<http://example.org/charlie#me> foaf:knows <http://example.org/alice#m
<http://example.org/charlie#me> foaf:name "Charlie" .
<http://example.org/alice#me> foaf:knows <http://example.org/snoopy> .
<http://example.org/snoopy> foaf:name "Snoopy"@en .
```

With SPARQL 1.1 one can query such graphs, load them into RDF stores and manipulate them in various ways.

§ 2. SPARQL 1.1 Query Language

Assuming the graph data from above is loaded into a SPARQL service (i.e., an HTTP service endpoint that can process SPARQL queries), the SPARQL 1.1 Query Language can be used to formulate queries ranging from simple graph pattern matching to complex queries. For

instance, one can ask using a SPARQL SELECT query for names of persons and the number of their friends:

```
PREFIX foaf: <http://xmlns.com/foaf/0.1/>
SELECT ?name (COUNT(?friend) AS ?count)
WHERE {
    ?person foaf:name ?name .
    ?person foaf:knows ?friend .
} GROUP BY ?person ?name
```

Just like in the earlier SPARQL1.0 specification [SPARQL-Query] from 2008, complex queries may include union, optional query parts, and filters; new features like value aggregation, path expressions, nested queries, etc. have been added in SPARQL 1.1. Apart from SELECT queries - which return variable bindings - SPARQL supports ASK queries - i.e. boolean "yes/no" queries - and CONSTRUCT queries - by which new RDF graphs can be constructed from a query result; all the new query language features of SPARQL 1.1 are likewise usable in ASK and CONSTRUCT queries.

Compared to SPARQL1.0, SPARQL 1.1 adds a number of new features to the query language, including subqueries, value assignment, path expressions, or aggregates - such as COUNT, as used in the above example query - etc.

The SPARQL 1.1 Query Language document defines the syntax and semantics of SPARQL 1.1 queries and provides various examples for their usage.

§ 3. Different query results formats supported by SPARQL 1.1 (XML, JSON, CSV, TSV)

Results of SELECT queries in SPARQL comprise bags of mappings from variables to RDF terms, often conveniently represented in tabular form. For instance, the query from Section 2 has the following results:

?name	?count
"Alice"	3

"Bob" 1

"Charlie" 1

In order to exchange these results in machine-readable form, SPARQL supports four common exchange formats, namely the Extensible Markup Language (XML), the JavaScript Object Notation (JSON), Comma Separated Values (CSV), and Tab Separated Values (TSV). These results formats are described in three different documents:

- the SPARQL Query Results XML Format (*please, particularly note that the SPARQL 1.1 WG has made some minor Errata to this specification*),
- the SPARQL 1.1 Query Results JSON Format, and
- the SPARQL 1.1 Query Results CSV and TSV Formats

These documents specify details of how particular solutions and RDF terms occurring in solutions are encoded in the respective target formats.

The results of our example query, in these three formats look as follows.

XML:

```
<?xml version="1.0"?>
<sparql xmlns="http://www.w3.org/2005/sparql-results#">
 <head>
   <variable name="name"/>
   <variable name="count"/>
 </head>
 <results>
   <result>
     <binding name="name">
       <literal>Alice</literal>
     </binding>
     <binding name="count">
       <literal datatype="http://www.w3.org/2001/XMLSchema#integer">3<
     </binding>
   </result>
   <result>
     <binding name="name">
       <literal>Bob</literal>
```

```
      </binding>
      <binding name="count">
        <literal datatype="http://www.w3.org/2001/XMLSchema#integer">1<
      </binding>
    </result>
    <result>
      <binding name="name">
        <literal>Charlie</literal>
      </binding>
      <binding name="count">
        <literal datatype="http://www.w3.org/2001/XMLSchema#integer">1<
      </binding>
    </result>
  </results>
</sparql>
```

JSON:

```
{
  "head": {
    "vars": [ "name" , "count" ]
  } ,
  "results": {
    "bindings": [
      {
        "name": { "type": "literal" , "value": "Alice" } ,
        "count": { "datatype": "http://www.w3.org/2001/XMLSchema#integ
      } ,
      {
        "name": { "type": "literal" , "value": "Bob" } ,
        "count": { "datatype": "http://www.w3.org/2001/XMLSchema#integ
      } ,
      {
        "name": { "type": "literal" , "value": "Charlie" } ,
        "count": { "datatype": "http://www.w3.org/2001/XMLSchema#integ
      }
    ]
  }
}
```

CSV:

```
name,count
Alice,3
Bob,1
Charlie,1
```

TSV:

```
?name<TAB>?count
"Alice"<TAB>3
"Bob"<TAB>1
"Charlie"<TAB>1
```

(***Note:*** *tab characters are visually marked with '*`/<TAB>`*' here for illustration only.*)

§ 4. SPARQL 1.1 Federated Query

The SPARQL 1.1 Federated Query document describes an extension of the basic SPARQL 1.1 Query Language to explicitly delegate certain subqueries to different SPARQL endpoints.

For instance, in our example, one may want to know whether there is anyone among Alice's friends with the same name as the resource identified by the IRI <http://dbpedia.org/resource/Snoopy> at DBpedia. This can be done by combining a query for the names of friends with a remote call to the SPARQL endpoint at `/http://dbpedia.org/sparql` finding out the name of <http://dbpedia.org/resource/Snoopy> using the `/SERVICE` keyword as follows:

```
PREFIX foaf: <http://xmlns.com/foaf/0.1/>
SELECT ?name
WHERE {
    <http://example.org/alice#me> foaf:knows [ foaf:name ?name ] .
    SERVICE <http://dbpedia.org/sparql> { <http://dbpedia.org/resource
}
```

with the following result:

?name
"Snoopy"@en

Here, the first part of the pattern in the `/WHERE` part is still matched against the local SPARQL service, whereas the evaluation of the pattern following the `/SERVICE` keyword is delegated to the respective remote SPARQL service.

§ 5. SPARQL 1.1 Entailment Regimes

SPARQL could be used together with ontological information in the form of, for example, RDF Schema or OWL axioms. For instance, let us assume that - apart from the data about Alice - some ontological information in the form of RDF Schema [RDF-Schema] and OWL [OWL2-Overview] constructs defining the FOAF vocabulary is loaded into our example SPARQL service.

The FOAF ontology: (only an excerpt given)

```
@prefix foaf: <http://xmlns.com/foaf/0.1/> .
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
...
foaf:name rdfs:subPropertyOf rdfs:label .
...
```

The following query asks for labels of persons:

```
SELECT ?label
WHERE { ?person rdfs:label ?label }
```

A SPARQL engine that does not consider any special entailment regimes (on top of standard simple entailment) would not return any results for this query, whereas an RDF Schema aware query engine will return

?label

"Alice"

"Bob"

"Charlie"

"Snoopy"@en

since `/foaf:name` is a sub-property of `/rdfs:label`.

The SPARQL 1.1 Entailment Regimes specification defines which answers should be given under which entailment regime, specifying entailment regimes for RDF, RDF Schema, D-Entailment [RDF-MT], OWL [OWL2-Overview], and RIF [RIF-Overview].

§ 6. SPARQL 1.1 Update Language

The SPARQL 1.1 Update specification defines the syntax and semantics of SPARQL 1.1 update requests and provides various examples for their usage. Update operations can consist of several sequential requests and are performed on a collection of graphs in a Graph Store. Operations are provided to update, create and remove RDF graphs in a Graph Store.

For instance, the following request inserts a new friend of Alice named Dorothy into the default graph of our example SPARQL service and thereafter deletes all names of Alice's friends with an English language tag.

```
PREFIX foaf: <http://xmlns.com/foaf/0.1/> .

INSERT DATA { <http://www.example.org/alice#me> foaf:knows [ foaf:name
DELETE { ?person foaf:name ?mbox }
WHERE { <http://www.example.org/alice#me> foaf:knows ?person .
        ?person foaf:name ?name FILTER ( lang(?name) = "EN" ) .}
```

As the second operation shows, insertions and deletions can be dependent on the results of queries to the Graph Store; the respective syntax used in the WHERE part is derived from the SPARQL 1.1 Query Language.

§ 7. SPARQL 1.1 Protocol for RDF

The SPARQL 1.1 Protocol for RDF defines how to transfer SPARQL 1.1 queries and update requests to a SPARQL service via HTTP. It also defines how to map requests to HTTP GET and POST operations and what respective HTTP responses to such requests should look like.

For instance, the query from Section 3 above issued against a SPARQL query service hosted at `/http://www.example.org/sparql/` could according to this specification be wrapped into an HTTP GET request (where the query string is URI-encoded):

```
      GET /sparql/?query=PREFIX%20foaf%3A%20%3Chttp%3A%2F%2Fxmlns.com%
Host: www.example.org
User-agent: my-sparql-client/0.1
```

Details about response encoding and different operations for query and update requests, as well as supported HTTP methods, are described in the Protocol specification.

§ 8. SPARQL 1.1 Service Description

The SPARQL 1.1 Service Description document describes a method for discovering and an RDF vocabulary for describing SPARQL services made available via the SPARQL 1.1 Protocol for RDF.

According to this specification, a service endpoint, when accessed via an HTTP GET operation without further (query or update request) parameters should return an RDF description of the service provided. For instance, the following HTTP request:

```
GET /sparql/ HTTP/1.1
Host: www.example.org
```

issued against the SPARQL endpoint hosted at `/http://www.example.org/sparql/` should return an RDF description, using the Service Description vocabulary. Such a description

provides, for instance, information about the default dataset of the respective endpoint, or about SPARQL query language features and entailment regimes that are supported.

§ 9. SPARQL 1.1 Graph Store HTTP Protocol

For many applications and services that deal with RDF data, the full SPARQL 1.1 Update language might not be required. To this end, the SPARQL 1.1 Graph Store HTTP Protocol provides means to perform certain operations to manage collections of graphs directly via HTTP operations.

For instance, the first part of the update request in Section 4 above is a simple insertion of triples into an RDF graph. On a service supporting this protocol, such insertion can - instead of via a SPARQL 1.1 update request - directly be performed via an HTTP POST operation taking the RDF triples to be inserted as payload:

```
POST /rdf-graphs/service?graph=http%3A%2F%2Fwww.example.org%2Falice HT
Host: example.org
Content-Type: text/turtle
@prefix foaf: <http://xmlns.com/foaf/0.1/> .
<http://www.example.org/alice#me> foaf:knows [ foaf:name "Dorothy" ] .
```

Other direct HTTP operations for modifying (e.g. to use HTTP PUT to replace an entire graph, or HTTP DELETE to drop an RDF graph) or retrieving (via HTTP GET) RDF graphs are described in the SPARQL 1.1 Graph Store HTTP Protocol specification, which can be viewed as a lightweight alternative to the SPARQL 1.1 protocol in combination with the full SPARQL 1.1 Query and SPARQL 1.1 Update languages.

§ 10. Acknowledgements

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- Andy Seaborne, The Apache Software Foundation
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§ 12. Privacy Considerations

TODO

§ 13. Security Considerations

TODO

§ 14. Internationalization Considerations

TODO

A. Change Log

This section is non-normative.

TODO

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